Regex Matching

```
>>> import re

>>> re.fullmatch(pattern, string)
```

regex matching function

regex pattern

string to be checked.

→ returns bool

→ True if pattern match, else False.

	Character / Pattern		Meaning	Example	
1		r'abc'	Exact match	'abc'	
2	.	r'a.c'	any single character	'rac' 'rxc'	'rbc' 'ryc'
3	*	r'a*'	0 or more of the preceding character	'' (empty) 'aaa'	'a' 'aaaaaa'
4	+	r'a+'	1 or more of the preceding character	'a'	'aaa'
5	?	r'ab?'	0 or 1 of the preceding character	'a'	'ab'
6	[...]	r'[abc]' r'[0-9]' r[A-Za-z]	character set → matches ~~one~~ 1 char that is either 1 of the inside [...]	'a' 'b' 'c' '0' '2' '8' 'A' 'B' 'c' 'd'	
7	[^...]	r'[^0-9]	Not one of these → ~~not~~ 1 char not of the inside [...]	'a' 'b' 'X' ~~'1'~~ ~~'8'~~	
8	{n} or {min, max}	r"[A-Za-z]{3}" r"[a-z]{2-5}"	Count repetitions — exact ← a range — 2 to 5 letters	'aBX' ~~'AB'~~ ~~'a'~~	'cdy' 'abc' 'xyza'

	Pattern/usage		Meaning	Example
(9)	\d	r'\d'	→ any digit → same as [0-9]	'1' '2' '0'
(10)	\w	r'\w'	any letter, number or _ → [A-Za-z0-9_]	'a' 'A' '1' '_'
(11)	\s	r'\s'	Any white space → or tab → or newline	' ' ' ' ' '

You can combine various patterns and characters to form complex patterns.

More matches

>>> re.search(pattern, target)

checks if pattern appears anywhere in the target.

>>> re.match(patt, tar)

checks only from the start.

Matching special characters

→ use '\' which is backslash.

eg. r'\.+' → '.' or '....' or '.....'

r'a\+b' → 'a+b'

Grouping: using ( )

r'(abc)' → matches 'abc' as a single unit.

r'(Mr|Ms)' → matches Mr & Mas

r'(ab)+' → matches repetitions → 'ab' or 'ababab'

r(Cat|Dog|Bat) → matches → 'Cat' → 'Dog' → 'Bat' only